

6 - BGP

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Prerequisites:

My ID is **18** from the txt file.

AS Numbers assigned:

- **AS650181**
- **AS650182**
- **AS650183**

Loopback subnets (/28 each):

- AS650181: 10.18.1.0/28 → Loopback: 10.18.1.1/32
- AS650182: 10.18.2.0/28 → Loopback: 10.18.2.1/32
- AS650183: 10.18.3.0/28 → Loopback: 10.18.3.1/32

Transit networks (/30):

- Between AS1-AS2: 172.16.18.0/30
- Between AS2-AS3: 172.16.18.4/30
- Between AS3-AS1: 172.16.18.8/30

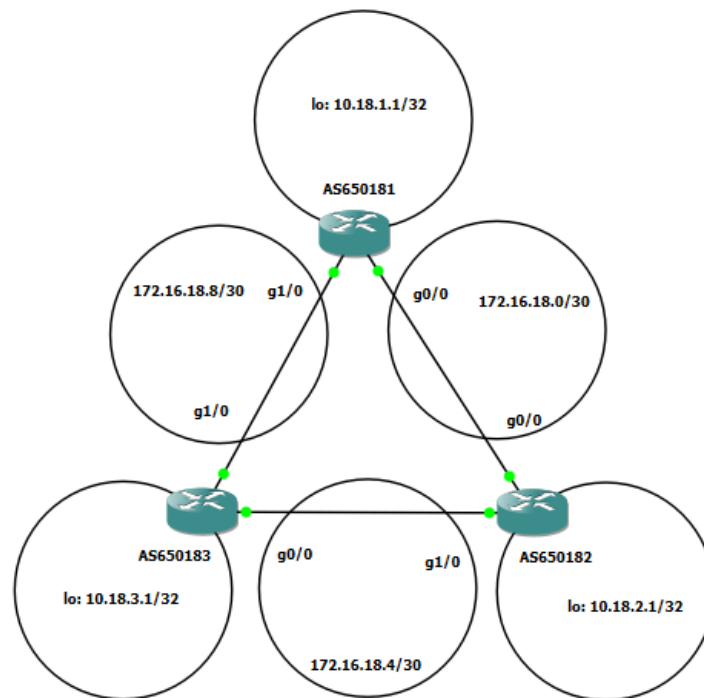


Figure. Network topology

Lab Tasks

Part A: Basic Router Configuration

Configure three routers in GNS3/EVE-NG, each representing a different AS.
For each router:

1. Configure hostname according to its AS number
 - I used CISCO 7200 as my routing solution with to GE adapters.
 - After creating a topology I did a snapshot

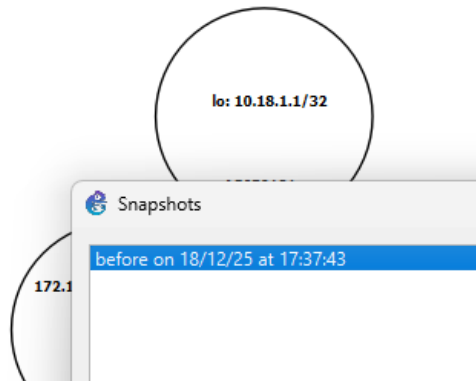
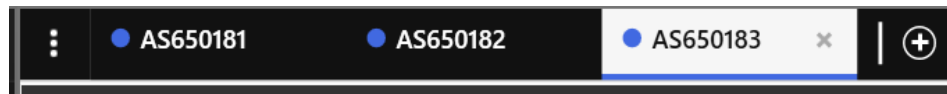


Figure. Snapshot

2. Configure loopback interface with the assigned /32 address

- I setup loopback interfaces (R1,R2,R3)



```
AS650181#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AS650181(config)#interface lo/
AS650181(config)#interface lo/
AS650181(config)#interface Loopback0
AS650181(config-if)#ip
*Dec 18 14:42:08.471: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
AS650181(config-if)#ip address 10.18.1.1 255.255.255.255
AS650181(config-if)#no shutdown
```

<pre>AS650182#conf t Enter configuration commands, one per line. End with CNTL/Z. AS650182(config)#interface Loopback0 AS650182(config-if)# *Dec 18 14:42:58.687: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up AS650182(config-if)#ip address 10.18.2.1 255.255.255.255 AS650182(config-if)#no shutdown</pre>	<pre>AS650183#conf t Enter configuration commands, one per line. End with CNTL/Z. AS650183(config)#interface Loopback0 AS650183(config-if)#ip *Dec 18 14:43:30.763: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up AS650183(config-if)#ip address 10.18.3.1 255.255.255.255 AS650183(config-if)#no shut</pre>
---	---

Figures. loopback interfaces (R1 up,R2 left,R3 right)

3. Configure physical interfaces with appropriate IP addresses from the /30 transit networks

- Setup physical interface configuration with static IP addresses inside transit networks (I am left with 2 possible addresses for each node: 1,2 ; 5,6 ; 9,10)

```
AS650181(config-if)#interface g0/0
AS650181(config-if)#ip address 172.16.18.1 255.255.255.252
AS650181(config-if)#no shut
AS650181(config-if)#
*Dec 18 14:58:09.963: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Dec 18 14:58:10.963: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
AS650181(config-if)#interface g1/0
AS650181(config-if)#ip address 172.16.18.8 255.255.255.252
Bad mask /30 for address 172.16.18.8
AS650181(config-if)#ip address 172.16.18.10 255.255.255.252
AS650181(config-if)#no shut
```

```
AS650182(config-if)#interface g0/0
AS650182(config-if)#ip address 172.16.18.2 255.255.255.252
% Incomplete command.
AS650182(config-if)#ip address 172.16.18.2 255.255.255.252
AS650182(config-if)#no shut
AS650182(config-if)#
*Dec 18 14:59:14.695: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Dec 18 14:59:15.695: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
AS650182(config-if)#interface g1/0
AS650182(config-if)#ip address 172.16.18.5 255.255.255.252
AS650182(config-if)#no shut

AS650183(config-if)#interface g0/0
AS650183(config-if)#ip address 172.16.18.6 255.255.255.252
AS650183(config-if)#no shut
AS650183(config-if)#interface g1/0
*Dec 18 15:04:08.711: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Dec 18 15:04:09.711: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
AS650183(config-if)#interface g1/0
AS650183(config-if)#ip address 172.16.18.9 255.255.255.252
AS650183(config-if)#no shut
```

Figures. gigabyte interfaces (R1 up,R2 left,R3 right)

4. Enable routing and ensure interfaces are active

- Ensured routing is enabled using

`show ip interface brief`

```
AS650181#show ip inter br
Interface                IP-Address      OK? Method Status      Protoco
Ethernet0/0              unassigned      YES unset  administratively down  down
GigabitEthernet0/0       172.16.18.1     YES manual up          up
GigabitEthernet1/0       172.16.18.10    YES manual up          up
Loopback0                 10.18.1.1       YES manual up          up
```

Oct 10 23:02:20 2019: R2: 2- com10_11 configured from console by console

AS650182#show ip inter br	Interface	IP-Address	OK?	Method	Status	Protocol
	Ethernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet0/0	172.16.18.2	YES	manual	up	up
	GigabitEthernet1/0	172.16.18.5	YES	manual	up	up
	Loopback0	10.18.2.1	YES	manual	up	up

AS650183#show ip inter br	Interface	IP-Address	OK?	Method	Status	Protocol
	Ethernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet0/0	172.16.18.6	YES	manual	up	up
	GigabitEthernet1/0	172.16.18.9	YES	manual	up	up
	Loopback0	10.18.3.1	YES	manual	up	up

Figures. interfaces (R1 up,R2 left,R3 right)

- Ensured pinging between nodes inside each local net, e.g. router 3 can ping router 1 and 2 inside their nets (172.16.18.8/30 and 172.16.18.4/30) and can not inside different nets

```

AS650183#ping 172.16.18.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.18.10, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 8/18/40 ms
AS650183#ping 172.16.18.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.18.6, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
AS650183#ping 172.16.18.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.18.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
AS650183#ping 172.16.18.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.18.5, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 20/24/36 ms
AS650183#ping 172.16.18.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.18.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

```

Figure. R3 pinging neighbours and itself on nets

Part B: BGP Configuration

Configure BGP on all three routers:

1. Enable BGP routing with the correct AS number for each router

```
AS650182(config)#router bgp 650182
AS650182(config-router)#bgp log-neighbor-changes
```

```
AS650181#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AS650181(config)#router bgp 650181
AS650181(config-router)#bgp log-neighbor-changes
```

```
AS650182#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AS650182(config)#router bgp 650182
AS650182(config-router)#bgp log-neighbor-changes
AS650183#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AS650183(config)#router bgp 650183
AS650183(config-router)#bgp log-neighbor-changes
^
% Invalid input detected at '^' marker.
AS650183(config-router)#bgp log-neighbor-changes
```

Figures. BGP routing with AS number (R1 up,R2 left,R3 right)

2. Set router ID to match the loopback address

```
AS650182(config)#router bgp 650182
AS650182(config-router)# bgp router-id 10.18.2.1
```

```
AS650181(config-router)#bgp router-id 10.18.1.1
```

```
AS650182#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AS650182(config)#router bgp 650182
AS650182(config-router)#bgp router-id 10.18.2.1
```

```
AS650183(config-router)#bgp router-id 10.18.3.1
```

Figure. (R1,R2 ,R3)

3. Configure eBGP neighbors pointing to the other routers' interface IPs; They are up.

```
AS650182(config)#router bgp 650182
AS650182(config-router)# neighbor 172.16.18.1 remote-as 650181
AS650182(config-router)# neighbor 172.16.18.6 remote-as 650183
```

```
AS650181(config-router)#neighbor 172.16.18.2 remote-as 650182
AS650181(config-router)#neighbor 172.16.18.2 remote-as 650183
*Dec 18 15:20:42.415: %BGP-5-ADJCHANGE: neighbor 172.16.18.2 Up
AS650181(config-router)#neighbor 172.16.18.9 remote-as 650183
```

```
AS650182(config-router)#neighbor 172.16.18.1 remote-as 650181
AS650182(config-router)#neighbor 172.16.18.6 remote-as 650183
```

```
AS650183(config-router)#neighbor 172.16.18.10 remote-as 650181
AS650183(config-router)#neighbor 172.16.18.5 remote-as 650182
AS650183(config-router)#
*Dec 18 15:22:02.027: %BGP-5-ADJCHANGE: neighbor 172.16.18.10 Up
AS650183(config-router)#
*Dec 18 15:22:05.323: %BGP-5-ADJCHANGE: neighbor 172.16.18.5 Up
AS650183(config-router)#
```

Figure. (R1 R2 R3)

4. Activate BGP for IPv4 unicast on all neighbor relationships

```
router bgp {AS_ID}
address-family ipv4
neighbor {IP} activate
exit-address-family
```

```
AS650181(config-router)#address-family ipv4
AS650181(config-router-af)#neighbor 172.16.18.9 activate
AS650181(config-router-af)#neighbor 172.16.18.2 activate
AS650181(config-router-af)#exit-address-fam
```

```

AS650182(config-router)#address-family ipv4
AS650182(config-router-af)#neighbor 172.16.18.1 activate
AS650182(config-router-af)#neighbor 172.16.18.6 activate
AS650182(config-router-af)#exit-add-fam
                                     ^
% Invalid input detected at '^' marker.

AS650182(config-router-af)#exit-addr
AS650182(config-router-af)#exit-address-family
AS650182(config-router)#

```

```

AS650183(config-router)#address-family ipv4
AS650183(config-router-af)#neighbor 172.16.18.5 activate
AS650183(config-router-af)#neighbor 172.16.18.10 activate
AS650183(config-router-af)#exit-address-family
AS650183(config-router)#

```

Figure. (R1 R2 R3)

5. Advertise the /28 loopback subnet and the /32 loopback address via BGP network statements

```

AS650181(config-router)#network 10.18.1.0 mask 255.255.255.240
AS650181(config-router)#network 10.18.1.1 mask 255.255.255.255

```

```

AS650181(config-router)#network 10.18.1.0 mask 255.255.255.240
AS650181(config-router)#network 10.18.1.1 mask 255.255.255.255
AS650181(config-router)#
AS650182(config-router)#network 10.18.2.0 mask 255.255.255.240
AS650182(config-router)#network 10.18.2.1 mask 255.255.255.255
AS650182(config-router)#
AS650183(config-router)#network 10.18.3.0 mask 255.255.255.240
AS650183(config-router)#network 10.18.3.1 mask 255.255.255.255
AS650183(config-router)#

```

Figure. (R1 R2 R3)

6. Ensure that BGP is advertising networks that exist in the routing table

```
show ip route connected
```

```

AS650181#show ip route connected
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
C    10.18.1.1 is directly connected, Loopback0
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C    172.16.18.0/30 is directly connected, GigabitEthernet0/0
L    172.16.18.1/32 is directly connected, GigabitEthernet0/0
C    172.16.18.8/30 is directly connected, GigabitEthernet1/0
L    172.16.18.10/32 is directly connected, GigabitEthernet1/0

```

Figure. R1

```

AS650182#show ip route connected
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
C    10.18.2.1 is directly connected, Loopback0
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C    172.16.18.0/30 is directly connected, GigabitEthernet0/0
L    172.16.18.2/32 is directly connected, GigabitEthernet0/0
C    172.16.18.4/30 is directly connected, GigabitEthernet1/0
L    172.16.18.5/32 is directly connected, GigabitEthernet1/0

```

Figure. R2

```

AS650183#show ip route connected
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/32 is subnetted, 3 subnets
C       10.18.3.1 is directly connected, Loopback0
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.16.18.4/30 is directly connected, GigabitEthernet0/0
L       172.16.18.6/32 is directly connected, GigabitEthernet0/0
C       172.16.18.8/30 is directly connected, GigabitEthernet1/0
L       172.16.18.9/32 is directly connected, GigabitEthernet1/0
AS650183#
*Dec 18 15:33:43.911: %SYS-5-CONFIG_I: Configured from console by console

```

Figure. R3

Part C: Verification and Testing

Perform comprehensive verification of your configuration.

1. Check BGP neighbor establishment on all routers

Command: `show ip bgp summary`

Neighbours are present. State is established, prefixes>0

```

AS650181#show ip bgp summary
BGP router identifier 10.18.1.1, local AS number 650181
BGP table version is 4, main routing table version 4
3 network entries using 432 bytes of memory
5 path entries using 400 bytes of memory
5/3 BGP path/bestpath attribute entries using 720 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1648 total bytes of memory
BGP activity 3/0 prefixes, 5/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/Pfx
Rcd
172.16.18.2    4      650182     22     22       4    0    0 00:15:58    2
172.16.18.9    4      650183     21     20       4    0    0 00:14:38    2
AS650181#

```

Figure. R1

```

AS650182#show ip bgp summary
BGP router identifier 10.18.2.1, local AS number 650182
BGP table version is 4, main routing table version 4
3 network entries using 432 bytes of memory
5 path entries using 400 bytes of memory
5/3 BGP path/bestpath attribute entries using 720 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1648 total bytes of memory
BGP activity 3/0 prefixes, 5/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/Pfx
Rcd
172.16.18.1    4      650181    22     22       4    0   0 00:15:57    2
172.16.18.6    4      650183    20     19       4    0   0 00:14:34    2

```

Figure. R2

```

AS650183#show ip bgp summary
BGP router identifier 10.18.3.1, local AS number 650183
BGP table version is 4, main routing table version 4
3 network entries using 432 bytes of memory
5 path entries using 400 bytes of memory
5/3 BGP path/bestpath attribute entries using 720 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1648 total bytes of memory
BGP activity 3/0 prefixes, 5/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/Pfx
Rcd
172.16.18.5    4      650182    19     20       4    0   0 00:14:33    2
172.16.18.10   4      650181    20     21       4    0   0 00:14:36    2

```

Figure. R3

2. Examine the BGP table to verify route learning

I examine correct AS pathes and all 3 internal loopbacks present, hops are right, weight is seen for each loopback network

Command: `show ip bgp`

```

AS650181#show ip bgp
BGP table version is 4, local router ID is 10.18.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
*>  10.18.1.1/32   0.0.0.0             0         32768 i
*   10.18.2.1/32   172.16.18.9         0           0 650183 650182 i
*   10.18.2.1/32   172.16.18.2         0           0 650182 i
*   10.18.3.1/32   172.16.18.2         0           0 650182 650183 i
*>  10.18.3.1/32   172.16.18.9         0           0 650183 i

```

Figure. R1

```

AS650182#show ip bgp
BGP table version is 4, local router ID is 10.18.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
*   10.18.1.1/32   172.16.18.6             0           0 650183 650181 i
*>  10.18.1.1/32   172.16.18.1             0           0 650181 i
*>  10.18.2.1/32   0.0.0.0                 0          32768 i
*   10.18.3.1/32   172.16.18.1             0           0 650181 650183 i
*>  10.18.3.1/32   172.16.18.6             0           0 650183 i
AS650182#

```

Figure. R2

```

AS650183#show ip bgp
BGP table version is 4, local router ID is 10.18.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
*   10.18.1.1/32   172.16.18.5             0           0 650182 650181 i
*>  10.18.1.1/32   172.16.18.10            0           0 650181 i
*   10.18.2.1/32   172.16.18.10            0           0 650181 650182 i
*>  10.18.2.1/32   172.16.18.5             0           0 650182 i
*>  10.18.3.1/32   0.0.0.0                 0          32768 i
AS650183#

```

Figure. R3

3. Check the global routing table for BGP-learned routes

Command: `show ip route bgp`

BGP learned routes are present inside each node correspondingly.

For example for R3:

```

10.0.0.0/32 is subnetted, 3 subnets
B    10.18.1.1 [20/0] via 172.16.18.10
B    10.18.2.1 [20/0] via 172.16.18.5

```

```

AS650181#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
B       10.18.2.1 [20/0] via 172.16.18.2, 00:10:09
B       10.18.3.1 [20/0] via 172.16.18.9, 00:09:38
AS650181#

```

Figure. R1

```

AS650182#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
B       10.18.1.1 [20/0] via 172.16.18.1, 00:11:33
B       10.18.3.1 [20/0] via 172.16.18.6, 00:09:42
AS650182#

```

Figure. R2

```

AS650183#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
B       10.18.1.1 [20/0] via 172.16.18.10, 00:11:38
B       10.18.2.1 [20/0] via 172.16.18.5, 00:10:18
AS650183#

```

Figure. R3

4. Test connectivity between all loopback interfaces

Pinging to loopback neighbour networks is done:

```
AS650181#ping 10.18.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.18.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/9/12 ms
AS650181#ping 10.18.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.18.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/16/24 ms
AS650181#
```

Figure. R1

```
AS650182#ping 10.18.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.18.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/8/12 ms
AS650182#ping 10.18.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.18.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/9/12 ms
AS650182#
```

Figure. R2

```
AS650183#ping 10.18.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.18.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/20 ms
AS650183#ping 10.18.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.18.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/12 ms
AS650183#
```

Figure. R3

5. Perform traceroute between different ASes to verify the path taken

Traceroutes show correct hops and right addresses — topology match and no loops.

```
Success rate is 100 percent (5/5), round-trip r
AS650181#traceroute 10.18.3.1
Type escape sequence to abort.
Tracing the route to 10.18.3.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.18.9 8 msec 28 msec 12 msec
AS650181#traceroute 10.18.2.1
Type escape sequence to abort.
Tracing the route to 10.18.2.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.18.2 8 msec 20 msec 8 msec
AS650181#traceroute 10.18.1.1
Type escape sequence to abort.
Tracing the route to 10.18.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.18.1.1 4 msec 0 msec 0 msec
AS650181#
```

Figure. R1

```
AS650182#traceroute 10.18.1.1
Type escape sequence to abort.
Tracing the route to 10.18.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.18.1 8 msec 36 msec 20 msec
AS650182#traceroute 10.18.3.1
Type escape sequence to abort.
Tracing the route to 10.18.3.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.18.6 16 msec 4 msec 8 msec
AS650182#traceroute 10.18.2.1
Type escape sequence to abort.
Tracing the route to 10.18.2.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.18.2.1 4 msec 4 msec 0 msec
```

Figure. R2

```
AS650183#traceroute 10.18.3.1
Type escape sequence to abort.
Tracing the route to 10.18.3.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.18.3.1 0 msec 4 msec 0 msec
AS650183#traceroute 10.18.2.1
Type escape sequence to abort.
Tracing the route to 10.18.2.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.18.5 8 msec 16 msec 12 msec
AS650183#traceroute 10.18.1.1
Type escape sequence to abort.
Tracing the route to 10.18.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.18.10 12 msec 32 msec 12 msec
AS650183#
```

Figure. R3



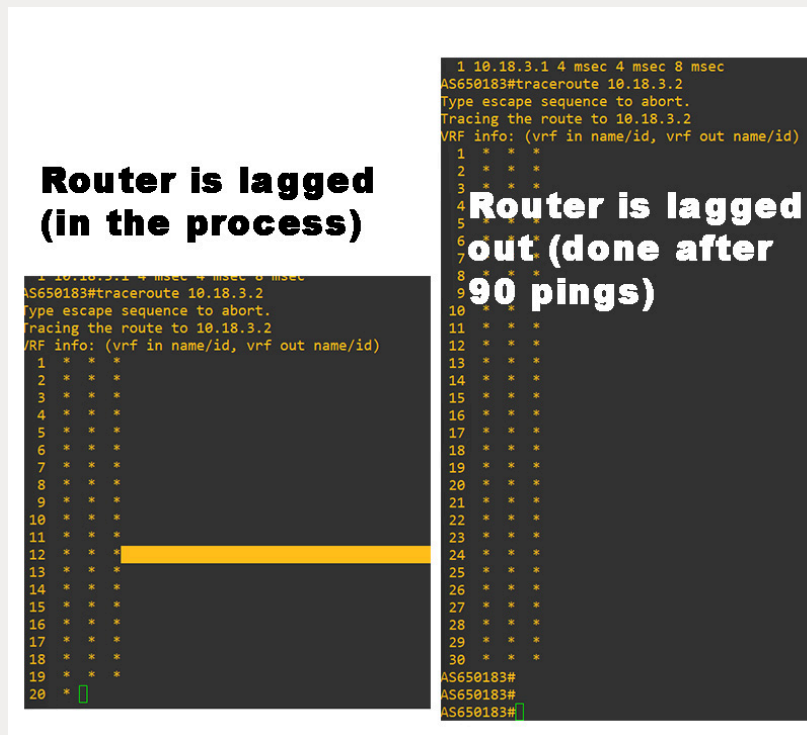
What can cause a crush on a router?

Writing different IP address for a traceroute, e.g. for R3 write traceroute 10.18.3.2 instead traceroute 10.18.3.1.

Figure bellow shows, that traceroute process is not stopping and any exit commands from console do not work — only one options is to manually reload it.

The second option is to wait, but what are limitations? $30 \times 3 \text{ pings} = 90 \text{ ping..}$

Figure. Lagging `traceroute`



Summary

Loopback interfaces are virtual and always up, so they are perfect for identifying a router. Even if a physical link goes down, the loopback address stays reachable. That's why loopbacks are used as router IDs and for basic connectivity tests.

Subnets define how IP addresses are divided and used. Small subnets like /30 are used for links between routers, /28 represents a small network owned by an AS, and /32 is used for a single address such as a loopback. Writing and understanding subnet masks is essential here. This helps avoid wasting IP addresses and keeps the network structured.

BGP is the protocol, that exchanges routing information between ASes. It advertises only nets, that already exist on the router and allows each AS to announce which IP address ranges it owns. Together, loopbacks, subnetting, and BGP make inter-AS routing stable and reliable technique.